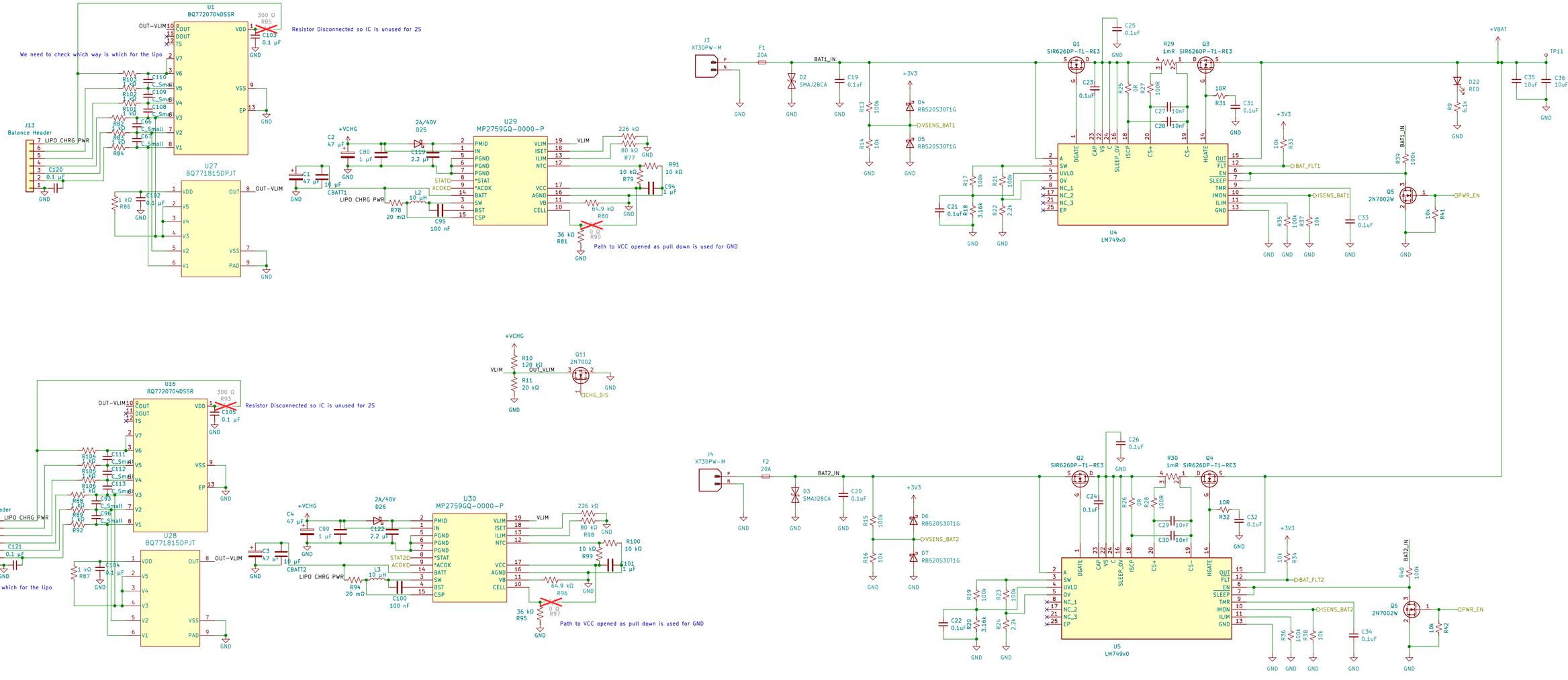

U1
BQ7720704DSSR
300 Ω
R85
C103
0.1 µF
OUT-VLIM
Resistor Disconnected so IC is unused for 2S
We need to check which way is which for the lipo
J13
Balance Header
7 LIPO CHRG PWR
R103
1 kΩ
R102
1 kΩ
R101
1 kΩ
R82
1 kΩ
R83
1 kΩ
R84
1 kΩ
C110
C_Small
C109
C_Small
C108
C_Small
C66
C_Small
C67
C_Small
C120
0.1 µF
U27
BQ771815DPJT
R86
1 kΩ
C102
0.1 µF
OUT-VLIM
GND
J3
XT30PW-M
F1
20A
BAT1_IN
D2
SMAJ28CA
C19
0.1uF
R13
100k
R14
10k
D4
RB520S30T1G
VSENS_BAT1
D5
RB520S30T1G
+3V3
C25
0.1uF
Q1
SIR626DP-T1-RE3
R29
1mR
Q3
SIR626DP-T1-RE3
C23
0.1uF
R25
0R
R27
100R
C27
10nF
C28
10nF
10R
R31
C31
0.1uF
+VBAT
TP11
D22
RED
R9
5.1k
C35
10uF
C36
10uF
+VCHG
C2
47 µF
C80
1 µF
2A/40V
D25
C119
2.2 µF
C1
47 µF
10 µF
CBATT1
LIPO CHRG PWR
L2
10 µH
R78
20 mΩ
C95
100 nF
STAT
ACOK
U29
MP2759GQ-0000-P
PMID
IN
PGND
PGND
PGND
*STAT
*ACOK
BATT
SW
BST
CSP
VLIM
ISET
ILIM
NTC
VCC
AGND
VB
CELL
226 kΩ
80 kΩ
R77
10 kΩ
R79
64.9 kΩ
R80
36 kΩ
R81
Path to VCC opened as pull down is used for GND
R91
10 kΩ
C94
1 µF
R17
100k
R21
100k
C21
0.1uF
R18
3.16k
R22
2.2k
U4
LM749x0
DGATE
CAP
VS
SLEEP_DV
ISCP
CS+
CS-
HGATE
SW
UVLO
OV
NC_1
NC_2
NC_3
EP
OUT
FLT
EN
SLEEP
TMR
IMON
ILIM
GND
10k
R33
BAT_FLT1
ISENS_BAT1
R35
100k
R37
10k
C33
0.1uF
BAT1_IN
R39
100k
Q5
2N7002W
PWR_EN
10k
R41
+VCHG
R10
120 kΩ
R11
20 kΩ
Q11
2N7002
VLIM
OUT_VLIM
QCHG_DIS
U16
BQ7720704DSSR
300 Ω
R93
C105
0.1 µF
OUT-VLIM
Resistor Disconnected so IC is unused for 2S
We need to check which way is which for the lipo
J14
Balance Header
7 LIPO CHRG PWR
R104
1 kΩ
R105
1 kΩ
R106
1 kΩ
R88
1 kΩ
R89
1 kΩ
R92
1 kΩ
C111
C_Small
C112
C_Small
C113
C_Small
C93
C_Small
C96
C_Small
C121
0.1 µF
U28
BQ771815DPJT
R87
1 kΩ
C104
0.1 µF
OUT-VLIM
GND
J4
XT30PW-M
F2
20A
BAT2_IN
D3
SMAJ28CA
C20
0.1uF
R15
100k
R16
10k
D6
RB520S30T1G
VSENS_BAT2
D7
RB520S30T1G
+3V3
C26
0.1uF
Q2
SIR626DP-T1-RE3
R30
1mR
Q4
SIR626DP-T1-RE3
C24
0.1uF
R26
0R
R28
100R
C29
10nF
C30
10nF
10R
R32
C32
0.1uF
+VCHG
C4
47 µF
C99
1 µF
2A/40V
D26
C122
2.2 µF
C3
47 µF
10 µF
CBATT2
LIPO CHRG PWR
L3
10 µH
R94
20 mΩ
C100
100 nF
STAT2
ACOK2
U30
MP2759GQ-0000-P
PMID
IN
PGND
PGND
PGND
*STAT
*ACOK
BATT
SW
BST
CSP
VLIM
ISET
ILIM
NTC
VCC
AGND
VB
CELL
226 kΩ
80 kΩ
R98
10 kΩ
R99
64.9 kΩ
R96
36 kΩ
R95
Path to VCC opened as pull down is used for GND
R100
10 kΩ
C101
1 µF
R19
100k
R23
100k
C22
0.1uF
R20
3.16k
R24
2.2k
U5
LM749x0
DGATE
CAP
VS
SLEEP_DV
ISCP
CS+
CS-
HGATE
SW
UVLO
OV
NC_1
NC_2
NC_3
EP
OUT
FLT
EN
SLEEP
TMR
IMON
ILIM
GND
10k
R34
BAT_FLT2
ISENS_BAT2
R36
100k
R38
10k
C34
0.1uF
BAT2_IN
R40
100k
Q6
2N7002W
PWR_EN
10k
R42
Processor: See STM Application Note AN5419
Waterloo Rocketry
Sheet: /Power Input/
File: pwr_in.kicad_sch
Title: Canard Board
Size: A2
Date: 2026-03-19
Rev: A
KiCad E.D.A. 10.0.7
Id: 5/13